5G RAN

ViNR Feature Parameter Description

Issue Draft A

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Contents

1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 01 (2025-03-10).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added support for downlink scheduling optimization during handovers for ViNR UEs. For details, see: • **4.1.3.2 Downlink Scheduling Optimization During Handovers** • **4.2.1 Benefits** • **4.3.2.1 Prerequisite Functions** • **4.4.1.1 Data Preparation** • **4.4.1.2 Using MML Commands**	Modified parameter: Added the **VINR_DL_SCH_OPT_DUR_HO_SW** option to the **NRDUCellDlSch.** ***VinrDlSchSwitch*** parameter.	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes

  None

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following feature.

Feature ID	Feature Name	Chapter/Section
FOFD-031203	VoNR	**4 ViNR Functions** **5 New Calling**

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
ViNR functions	Supported only in SA networking	**4 ViNR Functions**
New Calling	Supported only in SA networking	**5 New Calling**

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
ViNR functions	Supported only in low frequency bands	**4 ViNR Functions**
New Calling	Supported only in low frequency bands	**5 New Calling**

3 Overview

With the improvement of network capabilities in the 5G era, video over NR (ViNR) is introduced based on voice over NR (VoNR). ViNR requires quality of service (QoS) flows with a 5G QoS Identifier (5QI) of 2 to be set up to carry video services in addition to QoS flows with a 5QI of 1 set up for VoNR to carry voice services. ViNR UEs are also called VoNR UEs performing video services.

IP multimedia subsystem data channels (IMS DCs) are introduced to the voice and video channels for VoNR and ViNR on IMS networks. They enable support for New Calling, allowing subscribers to exchange various types of multimedia, such as texts, images, and videos. This facilitates visualized, interactive, and immersive communication experiences.

For details about the functions related to VoNR, see *VoNR*. This document describes the functions related to ViNR and New Calling.

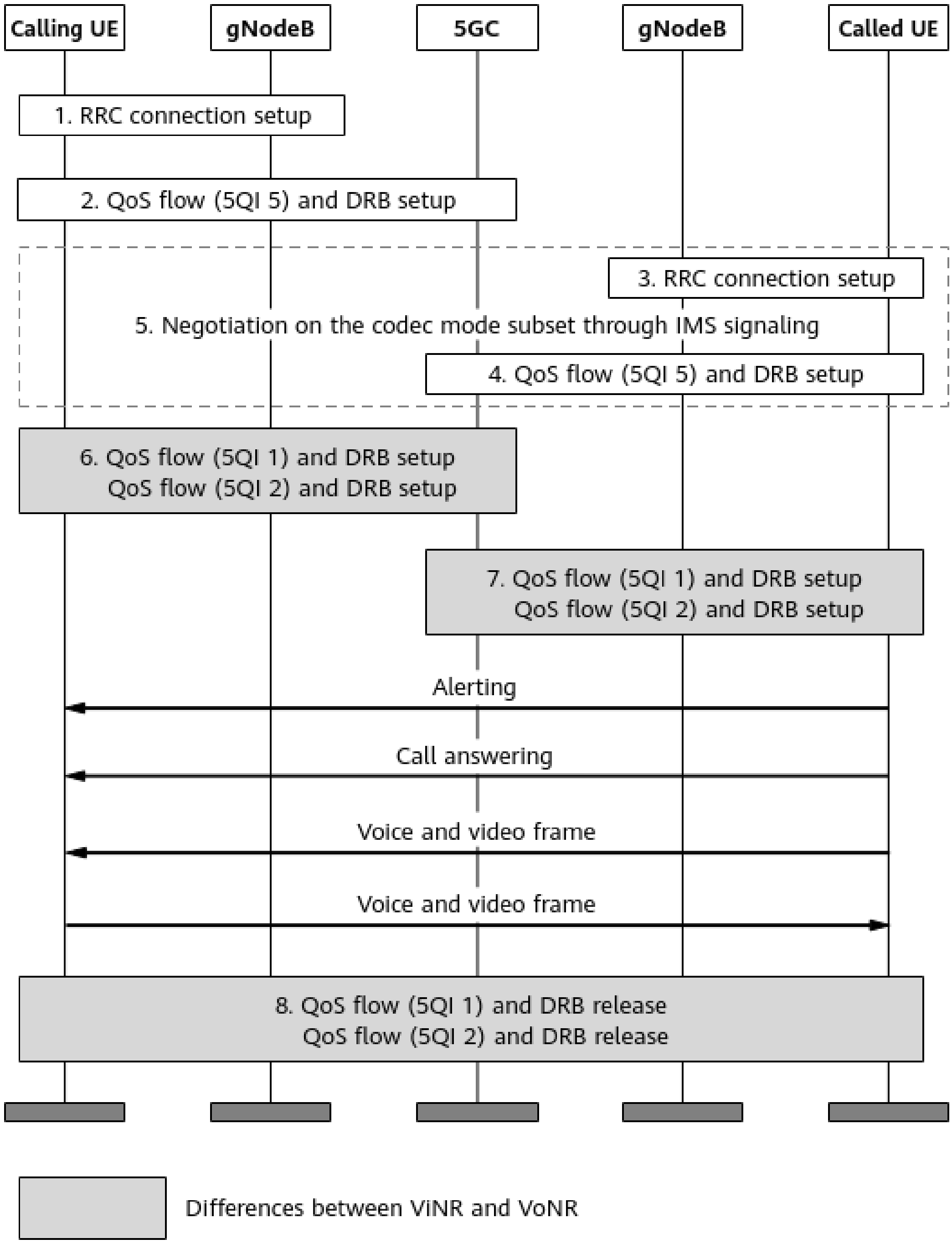
4 ViNR Functions

4.1 Principles

Basic ViNR functions are controlled by the **VONR_SW** option of the **NRCellAlgoSwitch**.***VonrSwitch*** parameter. When this option is selected, IP-based dedicated voice and video bearers are set up between UEs and the IMS on the NR network, enabling NR UEs to directly perform voice and video services on the NR network.

Figure 4-1 shows the process of setting up and releasing bearers between two UEs.

Figure 4-1 Voice and video bearer setup and release process



The detailed process is as follows:

1. When a UE initiates a call, an RRC connection is set up between the calling UE and its serving gNodeB.
2. The 5G core network (5GC) sets up a QoS flow with a 5QI of 5 for the calling UE to carry Session Initiation Protocol (SIP) signaling, and the gNodeB sets up the corresponding data radio bearer (DRB).
3. An RRC connection is set up between the UE that receives the call (called UE) and its serving gNodeB.

4. The 5GC sets up a QoS flow with a 5QI of 5 for the called UE to carry SIP signaling, and the gNodeB sets up the corresponding DRB.
5. The calling and called UEs and the IMS perform SIP negotiation on the codec scheme, IP address, port number, information of both calling and called UEs, and other voice service information.
6. After SIP negotiation is successful, the 5GC sets up a QoS flow with a 5QI of 1 to carry voice data flows and a QoS flow with a 5QI of 2 to carry video data flows for the calling UE, and the gNodeB sets up the corresponding DRBs.
7. The 5GC sets up a QoS flow with a 5QI of 1 to carry voice data flows and a QoS flow with a 5QI of 2 to carry video data flows for the called UE, and the gNodeB sets up the corresponding DRBs.
8. After the call ends, the calling and called UEs release their QoS flows with a 5QI of 1 and QoS flows with a 5QI of 2, and the gNodeBs release the corresponding DRBs. The default bearers with a 5QI of 5 are released only when the UEs enter the idle state.

As such, **in addition to QoS flows with a 5QI of 1 set up for VoNR, QoS flows with a 5QI of 2 also need to be set up for ViNR to carry video data flows.**

4.1.1 Coverage Improvement for ViNR

MAC CE-based rate adjustment, downlink coverage optimization, and enhanced uplink waveform adaptation are supported to improve uplink and downlink coverage performance for ViNR services.

4.1.1.1 MAC CE-based Rate Adjustment

Assume that the video coding rate of a UE does not match the air interface rate:

- When the air interface rate is high (that is, the network environment is good) but the video coding rate of the UE is low, only small video packets can be transmitted, leading to low resource usage. In this case, good network conditions are not fully utilized to improve the video service experience.
- When the air interface rate is low (that is, the network environment is poor) but the video coding rate of the UE is high, packet loss will occur during transmission, affecting the video service experience.

To address these issues, the gNodeB supports MAC CE-based rate adjustment to help UEs properly adjust video coding rates to cope with air interface rate changes. MAC CE stands for Media Access Control control element, which is used for control signaling at the MAC layer between the gNodeB and UE.

This function is enabled when the **ANBR_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected. The **gNBQciBearer**.***AnbrMode*** parameter specifies the MAC CE-based rate adjustment mode, which can be set to **UL_ONLY** (uplink only), **DL_ONLY** (downlink only), or **BOTH_SUPPORT** (both directions).

The function works as follows:

- Enables the gNodeB to provide information about recommended bit rates for UEs through a MAC CE based on the uplink or downlink air interface capability and load, as described in **Table 4-1**.

Table 4-1 gNodeB providing recommended bit rates for UEs based on the uplink or downlink air interface capability and load

gNodeB Detecting a UE's Air Interface Rate	gNodeB Notifying the UE of a Recommended Air Interface Rate Through a MAC CE	UE Making a Decision Based on the Recommended Air Interface Rate
The air interface rate is lower than a benchmark.	A low air interface rate is recommended.	Whether to decrease the video coding rate
The air interface rate is higher than a benchmark.	A high air interface rate is recommended.	Whether to increase the video coding rate

Figure 4-2 and **Figure 4-3** illustrate examples with the assumption that the benchmark is 2000 kbit/s, the recommended low air interface rate is 1200 kbit/s, and the recommended high air interface rate is 3000 kbit/s.

Figure 4-2 gNodeB helping a UE to decrease its rate

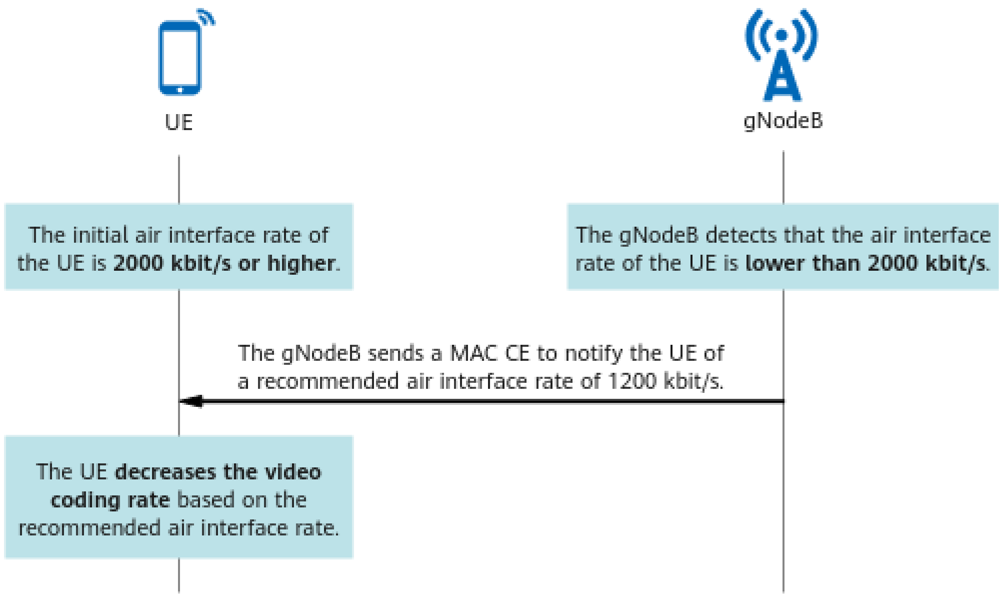
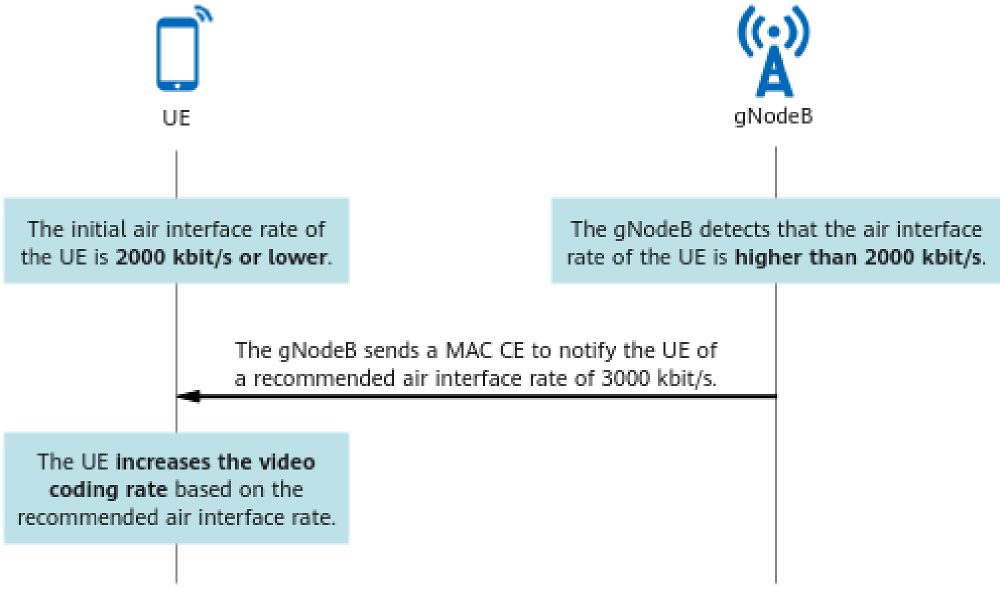


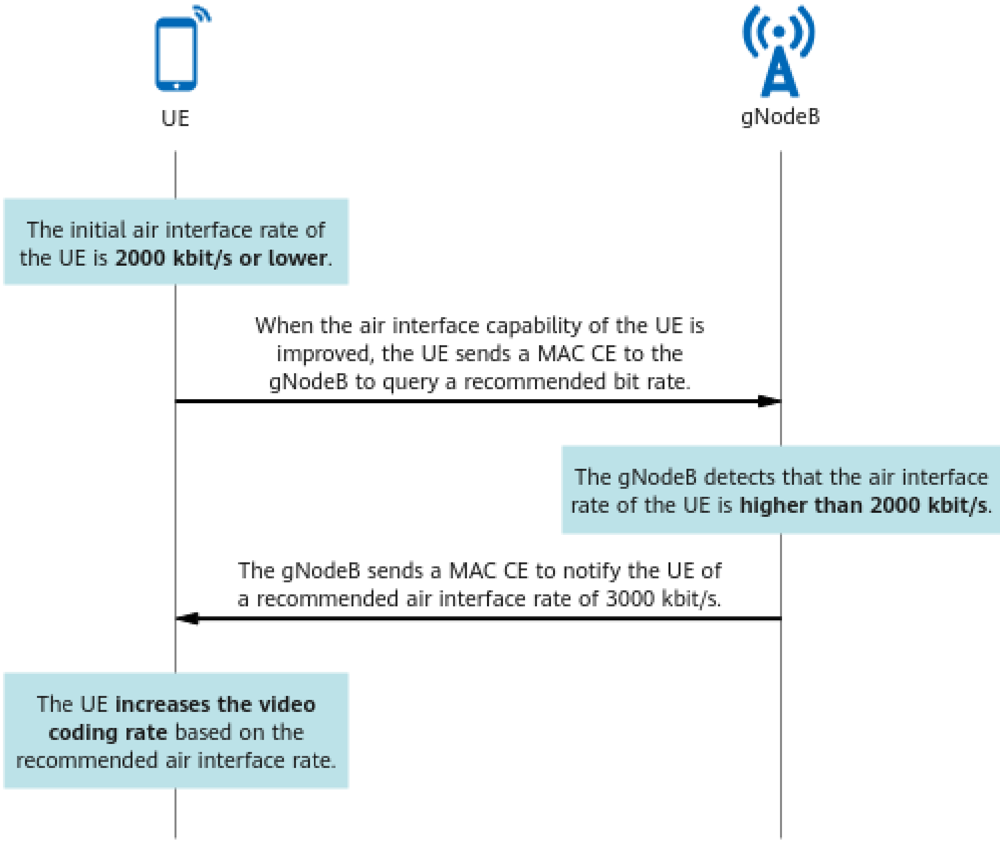
Figure 4-3 gNodeB helping a UE to increase its rate



- Allows UEs to query recommended bit rates provided by the gNodeB.

  A UE can send a MAC CE to the gNodeB to query a recommended bit rate. After detecting that the air interface rate of the UE is higher than a benchmark, the gNodeB sends a MAC CE to notify the UE of an increased recommended air interface rate. The UE then determines whether to increase the video coding rate based on this recommendation. **Figure 4-4** illustrates an example with the assumption that the benchmark is 2000 kbit/s and the recommended high air interface rate is 3000 kbit/s.

Figure 4-4 UE querying a recommended bit rate from the gNodeB to increase its rate



- Allows configuration of an interval at which the gNodeB provides recommended bit rates for a specific service type using the **gNBQciBearer**.***AnbrInterval*** parameter.

  A smaller value of this parameter results in more frequent transmission of recommended bit rates from the gNodeB to a UE, a higher signaling overhead, and greater timeliness of adaptive bit rate adjustment for user services. A larger value of this parameter results in the opposite effects.

The recommended air interface rates for this function are defined in 3GPP specifications. For details, see section 6.1.3.20 "Recommended bit rate MAC CE" in 3GPP TS 38.321 V15.5.0. The recommended low and high air interface rates in the preceding examples correspond to indexes 37 and 46, respectively, as listed in **Table 4-2**.

Table 4-2 Recommended bit rate values

Index	NR Recommended Bit Rate Value (kbit/s)
1	0
2	9
...	...
37	1200

Index	NR Recommended Bit Rate Value (kbit/s)
...	...
46	3000
...	...

For this function, the gNodeB only recommends video coding rates. The actual video coding rate is affected by factors such as the UE capability and is determined by the final negotiation between the calling and called UEs.

4.1.1.2 Downlink Coverage Optimization

Downlink coverage optimization for ViNR services involves downlink power compensation, increased downlink retransmissions, and downlink fixed rank-1 scheduling.

Downlink Power Compensation

Before scheduling ViNR services in the downlink, the gNodeB estimates the number of required RBs based on the volume of video services to be scheduled. If the estimated number is less than the total number of available RBs, the gNodeB will calculate the increase in the PDSCH power per RB based on the remaining power of the current cell.

This way, downlink power compensation is performed for ViNR UEs to improve their video service experience.

This function is enabled when the **DL_SERV_EXP_IMP_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

With this function, the maximum increase in the PDSCH power per RB depends on the **NRDUCellChnPwr**.***MaxPdschConvPwrOffset*** parameter value.

- If this parameter is set to a non-zero value, the maximum increase in the PDSCH power per RB is the value of this parameter.
- If this parameter is set to **0**, the maximum increase in the PDSCH power per RB is 3 dB.

Increased Downlink Retransmissions

When 5QI 2 bearers work in unacknowledged mode (UM), the gNodeB performs a maximum of four hybrid automatic repeat request (HARQ) retransmissions at the MAC layer for video service UEs. If a UE is located at the cell edge, four HARQ retransmissions may not be enough to ensure accurate downlink data transmission.

To address this issue, a maximum of eight HARQ retransmissions (nine HARQ transmissions) is allowed in the downlink to increase downlink retransmission opportunities, thereby increasing the downlink data transmission success rate under weak coverage.

This function is enabled when the **VINR_DL_SERV_EXP_IMP_SW** option of the **NRDUCellDlSch**.***VinrDlSchSwitch*** parameter is selected.

NOTE

When a 5QI 2 bearer works in acknowledged mode (AM) and four consecutive HARQ retransmissions fail, automatic repeat request (ARQ) retransmissions will be performed at the RLC layer up to 32 times. In this case, this function is not required.

You can run the **LST NRCELLQCIBEARER** command and check the value of **RLC Mode** in the command output to determine whether the current 5QI 2 bearer works in UM or AM.

Downlink Fixed Rank-1 Scheduling

If the downlink rank for ViNR UEs is high, the downlink video packet loss rate will increase. To reduce the downlink video packet loss rate and improve transmission reliability, rank 1 can be always used in downlink scheduling for ViNR UEs.

This function is enabled when the **VINR_DL_SERV_EXP_IMP_SW** option of the **NRDUCellDlSch**.***VinrDlSchSwitch*** parameter is selected.

4.1.1.3 Enhanced Uplink Waveform Adaptation

NR supports two waveforms: cyclic prefix-orthogonal frequency division multiplexing (CP-OFDM) and discrete Fourier transform-spread OFDM (DFT-s-OFDM).

- CP-OFDM allows for the use of non-contiguous frequency-domain resources and flexible resource allocation. It brings a high frequency-diversity gain but produces a high peak to average power ratio (PAPR).
- DFT-s-OFDM features a low PAPR (close to that of a single carrier) and high transmit power. However, it allows for the use of only contiguous frequency-domain resources.

By default, the gNodeB instructs the UE to always select CP-OFDM. **For ViNR UEs, using DFT-s-OFDM can improve uplink coverage.** Enhanced uplink waveform adaptation is enabled when the **RANK1_ADAPT_DFT_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter is selected.

Depending on RRC-reconfiguration-based or DCI-based dynamic waveform switching, this function further increases the probability of switching to DFT-s-OFDM and improves uplink coverage for ViNR UEs with one transmit antenna.

NOTE

For details about RRC-reconfiguration-based dynamic waveform switching, see *Turbo Uplink (Low-Frequency TDD)*.

For details about DCI-based dynamic waveform switching, see *Uplink Scheduling*.

4.1.2 Video Quality Improvement for ViNR

PDSCH HARQ retransmission optimization, PDCCH CCE aggregation level increase, optimized downlink MCS index selection, QCI-specific scheduling based on uplink and downlink target IBLERs, downlink RLC time-domain duplication, and downlink buffer jitter elimination are supported to improve ViNR service quality.

4.1.2.1 PDSCH HARQ Retransmission Optimization

A UE will report discontinuous transmission (DTX) information to the gNodeB if it does not receive the downlink control information (DCI) for PDSCH scheduling. If the gNodeB then mistakenly detects DTX as negative acknowledgment (NACK), it will use more RBs for retransmission, probably causing consecutive retransmission failures and video packet loss.

With PDSCH HARQ retransmission optimization, the HARQ retransmission mechanism is optimized when the HARQ feedback for video services is NACK.

When this function is enabled and both of the following conditions are met, the base station allocates the same modulation and coding scheme (MCS), number of symbols, rank, and number of RBs on the retransmission occasion as those on the initial transmission occasion to ensure that the transport block size (TBS) for retransmission is the same as that for initial transmission and uses more RBs for multiple retransmissions. This increases the probability of successful retransmission, improving downlink transmission reliability.

- The number of available symbols on the retransmission occasion is greater than or equal to that on the initial transmission occasion.
- The number of remaining RBs on the retransmission occasion is greater than or equal to that on the initial transmission occasion.

This function is enabled when the **DL_RETRANS_ADJ_OPT_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

4.1.2.2 PDCCH CCE Aggregation Level Increase

PDCCH control channel element (CCE) aggregation levels apply to both video and data services. When a low PDCCH CCE aggregation level is selected and video service UEs are located at the cell edge, the probability of DCI miss-detection increases. This causes downlink video packet loss and negatively affects video service experience.

This function increases the PDCCH CCE aggregation level used for retransmission of video service UEs by an offset.

The offset is specified by the **gNBDUQciAlgoParamGrp**.***RetransCceAggLevelOfs*** parameter. A larger value of this parameter results in a higher PDCCH CCE aggregation level used for retransmission of video service UEs, a lower probability of DCI miss-detection, and higher video transmission reliability. However, video services will occupy more CCE resources, causing fewer CCE resources available for data services.

4.1.2.3 Optimized Downlink MCS Index Selection for Initial Transmission

If high downlink MCS indexes are selected for the initial transmission to ViNR UEs, the downlink video packet loss rate will increase. As such, the downlink MCS indexes selected for the initial transmission to ViNR UEs must be appropriate to ensure the reliability of video service transmission.

Optimized downlink MCS index selection for initial transmission decreases the downlink MCS indexes for the initial transmission of video services to reduce the downlink video packet loss rate and improve video transmission reliability.

The amount to decrease individual downlink MCS indexes for the initial transmission of video services is specified by the **NRDUCellDlSch**.***VinrDlInitTransMcsDecrease*** parameter. A larger value of this parameter results in higher video transmission reliability but lower spectral efficiency and average downlink cell throughput.

4.1.2.4 QCI-specific Scheduling Based on Uplink and Downlink Target IBLERs

The functions of QCI-specific scheduling based on uplink and downlink target initial block error rates (IBLERs) perform outer loop link adaptation (OLLA) on services carried on bearers with a specific QCI, enabling the uplink and downlink IBLERs to converge on the configured uplink and downlink target IBLERs. With the functions, smaller target IBLERs can be configured to reduce the BLER, number of retransmissions, and delay over the air interface. For details, see *Service-differentiated Experience Guarantee*.

4.1.2.5 Downlink RLC Time-Domain Duplication

When the downlink packet loss rate of ViNR UEs is high, downlink video packets can be duplicated in the time domain to reduce their transmission failure rate and improve the downlink transmission reliability of video services.

This function enables downlink time-domain duplication at the RLC layer (in UM only) in a non-heavy-load cell when 5QI 2 bearers are set up, thereby reducing the packet loss rate.

This function is enabled when the **DL_RLC_TIME_DOM_DUPLICATION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected.

However, as this function causes a long transmission interval between the original video packet and a duplicated video packet, the PDCP reordering timer on the UE side may expire. To solve this problem, you are advised to set the **gNBPdcpParamGroup**.***UePdcpReorderingTimer*** parameter to **MS100**.

In addition, time-domain duplication upon downlink residual block errors can be enabled by selecting the **DL_RESID_ERR_TIME_DOM_DUP_SW** option of the **NRCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter. With this function, if downlink residual block errors occur in both the original and duplicated video packets, a video packet is additionally duplicated for transmission, further reducing the downlink packet loss rate.

NOTE

In a heavy-load cell, the downlink PRB usage reaches the threshold specified by the **NRDUCellDlSch**.***DlSchHeavyLoadPrbThld*** parameter (80% by default) or the CCE allocation failure rate is higher than 30%.

Upon consecutive packet loss, performing downlink RLC time-domain duplication will waste resources. In this case, exit from time-domain duplication upon consecutive packet loss can be used to reduce resource waste. This function is enabled when the **CONT_PKT_LOSS_EXIT_TIME_DUP_SW** option of the **NRDUCellAlgoSwitch**.***ServiceDiffSwitch*** parameter is selected. If a UE's video services experience a consecutive loss of downlink packets exceeding the threshold specified by the **NRDUCellVoiceAlgo**.***TimeDupExtContPktLossThld*** parameter, the base station stops downlink RLC time-domain duplication for the UE's video services.

4.1.2.6 Downlink Buffer Jitter Elimination

For a ViNR UE, when the uplink air interface is congested, a handover is performed, or the core network transmission delay changes, video packet congestion and jitter occur. Jitter indicates the changes in video packet receiving intervals.

With downlink buffer jitter elimination, the base station serving the target cell periodically sends video packets in sequence after buffering the 5QI 2 packets from the peer end to eliminate jitter and improve video quality.

This function takes effect when the **NRCellServExp**.***VinrDUitterEliminateDur*** parameter (specifying the downlink jitter elimination duration for ViNR) is set to a non-zero value for both the source and target cells. The base station buffers video frames, performs delay shaping on them, and sends them at an expected frame interval in the downlink. It sends the first video frame in a buffer queue to a UE when the video frame has been buffered for a jitter elimination duration or when the queue becomes full and a new video frame arrives.

4.1.3 Mobility Management for ViNR

4.1.3.1 Video-Quality-based Inter-Frequency Handover

If there are ViNR UEs with poor video quality on the network, this function allows video-quality-based inter-frequency handover for these UEs to ensure the quality of their video services.

This function is enabled when the **QLTY_BASED_INTER_FREQ_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

This function follows the same procedure as coverage-based inter-frequency handover. It supports measurement-based mode but not blind mode. The processes of handover function start decision, measurement configuration delivery, and measurement reporting are different from those in coverage-based inter-frequency handover. This section describes only the differences. For details about the coverage-based inter-frequency handover procedure, see *SA Mobility Management in Connected Mode*.

Handover Function Start Decision

After video-quality-based inter-frequency handover is enabled, the following operations apply within a service quality evaluation period (specified by the **gNBQciAlgoParamGrp**.***ServPlrEvalPeriod*** parameter):

- When the uplink or downlink packet loss rate of UE-specific video services is higher than the packet loss rate threshold for service-quality-based inter-frequency handover, the video quality of the UE deteriorates. In this case, video-quality-based inter-frequency measurement is triggered.
- When both the uplink and downlink packet loss rates of UE-specific video services are lower than or equal to the packet loss rate threshold for service quality recovery, the video quality of the UE improves. In this case, video-quality-based inter-frequency measurement is stopped.

The packet loss rate thresholds for service-quality-based inter-frequency handover and for service quality recovery are specified by the

gNBQciAlgoParamGrp.***ServQltyInterFreqHoPlrThld*** and **gNBQciAlgoParamGrp**.***ServQltyRecoveryPlrThld*** parameters, respectively.

Measurement Configuration Delivery

The gNodeB delivers the measurement configurations related to neighboring frequencies in descending order of frequency priority. That is, measurement configurations related to high-priority frequencies are preferentially delivered. The priority of a neighboring frequency is specified by the **NRCellFreqRelation**.***VonrQualityFreqPriority*** parameter.

- If this parameter is set to a value ranging from **0** to **16**, a larger value indicates a higher priority.
- If this parameter is set to **255**, the frequency will not support ViNR.

The gNodeB delivers measurement configurations for event A5 (the signal quality of the serving cell drops below threshold 1 and the signal quality of a neighboring cell exceeds threshold 2) related to neighboring frequencies. Event A5 supports only RSRP-based triggering. (RSRP stands for reference signal received power.) For details about event A5, see *SA Mobility Management in Connected Mode*.

In video-quality-based inter-frequency handover, threshold 1 for event A5 is fixed at –43 dBm, and threshold 2 for event A5 is specified by the **NRCellServExp**.***VonrQltyInterFA5RsrpThld2*** parameter.

Measurement Reporting

After receiving the measurement configurations delivered by the gNodeB, UEs send event A5 measurement reports if the entering condition for event A5 is met. For ViNR UEs, configurations of inter-frequency neighboring cell measurement vary with the settings of the interference flag of a neighboring frequency (specified by the **NRCellFreqRelation**.***FreqIntrfFlag*** parameter) according to the requirements of different scenarios on the live network. An RSRP, reference signal received quality (RSRQ), or signal to interference plus noise ratio (SINR) threshold is configured for the neighboring cells based on the corresponding parameter setting, as described in **Table 4-3**.

Table 4-3 Neighboring cell filtering rules and corresponding thresholds

Value of NRCellFreqRelation.*FreqIntrfFlag*	Neighboring Cell Filtering Rule	Threshold
NO_INTRF	Neighboring cells are filtered based only on RSRP.	RSRP: **NRCellServExp**.***VonrQltyInterFA5RsrpThld2***
INTRF	Neighboring cells are filtered based on RSRP and RSRQ.	• RSRP: **NRCellServExp**.***VonrQltyInterFA5RsrpThld2*** • RSRQ: **NRCellServExp**.***VonrQltyInterFA5RsrqThld***

Value of NRCellFreqRelation.*FreqIntrfFlag*	Neighboring Cell Filtering Rule	Threshold
INTRF_SINR	Neighboring cells are filtered based on RSRP and SINR.	● RSRP: **NRCellServExp**.***VonrQltyInterFA5RsrpThld2*** ● SINR: **NRCellServExp**.***VonrQltyInterFA5SinrThld***

When the measured RSRP, RSRQ, or SINR value of a neighboring cell reported by a UE is greater than the corresponding threshold, the event A5 measurement report is valid.

Video-Quality-based Redirection

If no neighbor relationship is configured between the serving cell of a ViNR UE and the neighboring cell with the strongest signals in the measurement report from this UE, or if the neighboring cell encounters physical cell identifier (PCI) confusion, the ViNR UE cannot be handed over to that neighboring cell during a video-quality-based inter-frequency handover. This will cause service drops or ping-pong handovers.

To reduce the probability of service drops or ping-pong handovers in the preceding handover scenarios, video-quality-based redirection enables the base station to select the frequency indicated in a measurement report from a UE for redirection.

This function is enabled when the **QLTY_BASED_REDI_SW** option of the **NRCellAlgoSwitch**.***VoiceStrategySwitch*** parameter is selected.

Collaboration with Other Inter-Frequency Handover Functions

If video-quality-based inter-frequency handover is enabled together with any of the following inter-frequency handover functions, ping-pong handovers may occur, which negatively affects the service experience of ViNR UEs.

- Frequency-priority-based inter-frequency handover or CA-UE-specific frequency-priority-based inter-frequency handover. In this case, inter-frequency handover collaboration can be enabled by selecting the **INTER_FREQ_HO_COLLABORATION_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter to prevent the base station from initiating frequency-priority-based inter-frequency handovers for ViNR UEs, thereby avoiding ping-pong handovers.
- Service-based inter-frequency handover. In this case, the anti-ping-pong function for service-based and video-quality-based inter-frequency handovers can be enabled by selecting the **VOICE_INTER_FREQ_HO_COLLAB_SW** option of the **NRCellAlgoSwitch**.***InterFreqHoSwitch*** parameter to prevent the base station from triggering service-based inter-frequency handovers for ViNR UEs that are handed over to the local cell based on video quality within 10s. This way, the number of inter-frequency ping-pong handovers reduces.

For details about frequency-priority-based and service-based inter-frequency handovers, see *SA Mobility Management in Connected Mode*. For details about CA-UE-specific frequency-priority-based inter-frequency handover, see *CA Experience Improvement*.

4.1.3.2 Downlink Scheduling Optimization During Handovers

During handovers, UEs cannot be scheduled, causing jitter when ViNR UEs receive video packets after being handed over. This may result in asynchronous audio and video packets, degrading video call quality.

This function delays handovers for ViNR UEs and sends downlink video packets during the delay period to reduce jitter in ViNR services during handovers. When enabled together with both downlink buffer jitter elimination and downlink handover delay optimization for VoNR UEs, this function can also address the issue of asynchronous audio and video packets.

This function is enabled when the **VINR_DL_SCH_OPT_DUR_HO_SW** option of the **NRDUCellDlSch**.***VinrDlSchSwitch*** parameter is selected.

4.1.4 Collaboration with Other Functions

4.1.4.1 Collaboration with MU-MIMO

Multi-user multiple-input multiple-output (MU-MIMO) implements the spatial multiplexing of time-frequency resources for multiple UEs in both uplink and downlink, increasing uplink and downlink cell capacity and improving user experience.

- In the uplink, ViNR UEs can be paired for PUSCH MU-MIMO only when the **UL_MU_MIMO_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter is selected.
- In the downlink, ViNR UEs can be paired for PDSCH MU-MIMO only when the **VINR_DL_MUMIMO_PROHIBIT_SW** option of the **NRDUCellDlMimo**.***DLMuMimoSchSupplementSw*** parameter is deselected.

For details about MU-MIMO functions, see *MIMO (TDD)*.

4.1.4.2 DRX Wakeup Protection

In discontinuous reception (DRX) scenarios, if the gNodeB falsely detects an uplink scheduling request (SR) and incorrectly detects the cyclic redundancy check (CRC) result for the uplink scheduling, it will mistakenly consider that the UE wakes up from sleep time and enters active time and then perform downlink scheduling for the UE. This causes downlink video packet loss. For details about DRX, see *DRX*.

DRX wakeup protection enables downlink scheduling for ViNR services only after reliable uplink scheduling is determined. This reduces the downlink packet loss rate of ViNR services.

This function is enabled when the **DRX_WAKEUP_PROTECT_SW** option of the **gNBDUQciAlgoParamGrp**.***QciAlgoSwitch*** parameter is selected.

4.2 Network Analysis

4.2.1 Benefits

After basic ViNR functions are enabled, NR UEs can directly perform voice and video services on the NR network. **Table 4-4** describes the benefits of different ViNR functions.

Table 4-4 Benefits of ViNR functions

Category	Function Name	Benefits
Coverage improvement for ViNR	MAC CE-based rate adjustment	Improves video service experience by flexibly adjusting video bit rates based on network conditions.
	Downlink coverage optimization	Improves downlink transmission quality and downlink coverage.
	Enhanced uplink waveform adaptation	Improves uplink coverage.
Video quality improvement for ViNR	PDSCH HARQ retransmission optimization	Reduces the downlink video packet loss rate and improves video service experience.
	PDCCH CCE aggregation level increase	
	Optimized downlink MCS index selection for initial transmission	
	QCI-specific scheduling based on uplink and downlink target IBLERs	
	Downlink RLC time-domain duplication	
	Downlink buffer jitter elimination	Reduces video freezing and improves video service experience.
Mobility management for ViNR	Video-quality-based inter-frequency handover	Improves video service quality and experience by triggering inter-frequency handovers when the video quality is poor. Reduces the video service drop rate through video-quality-based redirection.

Category	Function Name	Benefits
	Downlink scheduling optimization during handovers	Reduces the jitter caused by handover interruption for ViNR UEs. Reduces the time difference in audio and video synchronization.
Collaboration with other functions	Collaboration with MU-MIMO	Improves video service experience.
	DRX wakeup protection	Reduces the downlink video packet loss rate and improves video service experience.

4.2.2 Impacts

The impacts between this function and other functions are described in **4.3.2.3 Function Impacts**.

After basic ViNR functions are enabled, basic VoNR functions also take effect. For details about the network impacts of basic VoNR functions, see *VoNR*.

Voice, video, and data services share NR air interface resources. If the ViNR traffic volume increases, there will be fewer resources available for data services and their throughput will drop.

In addition to the impacts of basic ViNR functions, **Table 4-5** describes the impacts of different ViNR functions.

Table 4-5 Network impacts of ViNR functions

Category	Function Name	Impacts
Coverage improvement for ViNR	MAC CE-based rate adjustment	No additional impact
	Downlink coverage optimization	Downlink power compensation raises the downlink transmit power for ViNR UEs, which increases the interference of ViNR UEs to neighboring cells.
	Enhanced uplink waveform adaptation	Enhanced uplink waveform adaptation increases the probability of switching to DFT-S-OFDM when rank 1 is used in the uplink for ViNR UEs. For more information, see *Uplink Scheduling*.
Video quality improvement for ViNR	PDSCH HARQ retransmission optimization	No additional impact

Category	Function Name	Impacts
	PDCCH CCE aggregation level increase	PDCCH CCE aggregation level increase allows ViNR services to occupy more CCE resources and data services to occupy fewer CCE resources, decreasing the downlink cell traffic volume (**Downlink Traffic Volume (DU)**).
	Optimized downlink MCS index selection for initial transmission	The larger the decrease in the downlink MCS indexes for the initial transmission of video services, the lower the spectral efficiency and the average downlink cell throughput (**Cell Downlink Average Throughput (DU)**).
	QCI-specific scheduling based on uplink and downlink target IBLERs	For details about the network impacts of QCI-specific scheduling based on uplink and downlink target IBLERs, see *URLLC*.
	Downlink RLC time-domain duplication	An increased number of downlink video packets for ViNR UEs prolongs the total downlink transmission delay of services with a specific 5QI at the RLC layer in a cell (**N.RLC.DL.Cell5QI.Delay**). This function increases resource consumption, slightly raises the average downlink PRB usage of a cell, and causes a small increase in the traffic carried on 5QI 2 bearers. Time-domain duplication upon downlink residual block errors will exacerbate the preceding impacts. Exit from time-domain duplication upon consecutive packet loss reduces the probability that downlink RLC time-domain duplication takes effect.
	Downlink buffer jitter elimination	Buffering 5QI 2 packets at the base station side prolongs the total downlink transmission delay of services with a specific 5QI at the RLC layer in a cell (**N.RLC.DL.Cell5QI.Delay**). This may also cause fluctuations in the MCS indexes and a decrease in the number of RBs used for downlink scheduling of ViNR UEs.
Mobility management for ViNR	Video-quality-based inter-frequency handover	Video-quality-based inter-frequency handover increases the number of inter-frequency handovers.

Category	Function Name	Impacts
	Downlink scheduling optimization during handovers	Service drops may occur due to delayed delivery of handover commands.
Collaboration with other functions	Collaboration with MU-MIMO	For details about the impacts of MU-MIMO, see the descriptions in *MIMO (TDD)*.
	DRX wakeup protection	DRX wakeup protection increases the jitter.

4.3 Requirements

4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-031203	VoNR	NR0S00VONR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	RRC-reconfiguration-based dynamic waveform switching	**UL_WAVEFORM_ADAPT_SW** option of the **NRDUCellAlgoSwitch**.***AdaptiveEdgeExpEnhSwitch*** parameter	*Turbo Uplink (Low-Frequency TDD)*	RRC-reconfiguration-based or DCI-based dynamic waveform switching must be enabled before enhanced uplink waveform adaptation is enabled.
Low-frequency TDD	DCI-based dynamic waveform switching	**DYNAMIC_WAVEFORM_SWITCHING_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter	*Uplink Scheduling* *Massive MIMO Uplink Boosting (Low-Frequency TDD)*	RRC-reconfiguration-based or DCI-based dynamic waveform switching must be enabled before enhanced uplink waveform adaptation is enabled.
Low-frequency TDD	Downlink RLC time-domain duplication	**DL_RLC_TIME_DOM_DUPLICATION_SW** option of the **NRDUCellQciBearer**.***ServiceDiffAlgoSwitch*** parameter	*ViNR*	Time-domain duplication upon downlink residual block errors depends on downlink RLC time-domain duplication. Downlink RLC time-domain duplication is required for exit from time-domain duplication upon consecutive packet loss to take effect.
Low-frequency TDD	QoS Class Identifier	**NRDUCellQciBearer**.***Qci***	None	When downlink RLC time-domain duplication is enabled, the **NRDUCellQciBearer**.***Qci*** parameter must be set to **2**.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	QoS Class Identifier	**NRCellQciBearer**.***Qci***	None	When time-domain duplication upon downlink residual block errors is enabled, the **NRCellQciBearer**.***Qci*** parameter must be set to **2**.
Low-frequency TDD	Video-quality-based inter-frequency handover	**QLTY_BASED_INTER_FREQ_HO_SW** option of the **gNBQciAlgoParamGrp**.***QciAlgoSwitch*** parameter	*ViNR*	Video-quality-based inter-frequency handover is required for video-quality-based redirection to take effect.
Low-frequency TDD	ViNR DL Jitter Eliminate Duration	**NRCellServExp**.***VinrDlJitterEliminate-Dur***	*ViNR*	When downlink scheduling optimization during handovers for ViNR UEs is enabled, the **NRCellServExp**.***VinrDlJitterEliminateDur*** parameter must be set to a non-zero value.
Low-frequency TDD	Downlink handover delay optimization for VoNR UEs	**VONR_DL_SCH_OPT_DUR_HO_SW** option of the **NRDUCellDlSch**.***VoiceDlSchSwitch*** parameter	*VoNR*	Downlink scheduling optimization during handovers for ViNR UEs depends on downlink handover delay optimization for VoNR UEs.

4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	High-speed Railway Superior Experience	**NRDUCell**.***HighSpeedFlag*** set to **HIGH_SPEED**	*High Speed Mobility*	Enhanced uplink waveform adaptation (controlled by the **RANK1_ADAPT_DFT_SW** option of the **NRDUCellChnCovAlgo**.***UlCoverageAlgoSwitch*** parameter) cannot be enabled in high-speed cells.

4.3.2.3 Function Impacts

None

4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.4 Networking

SA networking is required.

4.3.5 Others

This function also has the following requirements:

- UE requirements: This function has the same requirements for UEs as VoNR. For details, see *VoNR*.
- Core network requirements: The 5QI 2 rate limit configured on the core network will affect video service quality and experience. Therefore, appropriate configuration is required based on the actual conditions.

4.4 Operation and Maintenance

4.4.1 Data Configuration

4.4.1.1 Data Preparation

Table 4-6, **Table 4-7**, **Table 4-8**, **Table 4-9**, and **Table 4-10** describe the parameters used for function activation.

Table 4-6 Parameters used for activation of basic ViNR functions

Parameter Name	Parameter ID	Option	Setting Notes
VoNR Switch	**NRCellAlgoSwitch.*VonrSwitch***	VONR_SW	Select this option.

Table 4-7 Parameters used for activation of coverage improvement for ViNR

Parameter Name	Parameter ID	Option	Setting Notes
Service Differentiation Algorithm Switch	**NRDUCellQciBearer.*ServiceDiffAlgoSwitch***	ANBR_SW	Select this option if MAC CE-based rate adjustment is required according to the network plan.
ANBR Mode	**gNBQciBearer.*AnbrMode***	None	Set this parameter to **UL_ONLY**, **DL_ONLY**, or **BOTH_SUPPORT** to enable rate adjustment in the uplink, downlink, or both directions, respectively, according to the network plan.
ANBR Interval	**gNBQciBearer.*AnbrInterval***	None	Retain the default value.

Parameter Name	Parameter ID	Option	Setting Notes
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp.*QciAlgoSwitch***	DL_SERV_EXP_IMP_SW	Select this option if downlink power compensation is required according to the network plan.
ViNR Downlink Scheduling Switch	**NRDUCellDlSch**.***VinrDlSchSwitch***	VINR_DL_SERV_EXP_IMP_SW	Select this option if increased downlink retransmissions and downlink fixed rank-1 scheduling are required according to the network plan.
UL Coverage Algorithm Switch	**NRDUCellChnCovAlgo.*UlCoverageAlgoSwitch***	RANK1_ADAPT_DFT_SW	Select this option if enhanced uplink waveform adaptation is required according to the network plan.

Table 4-8 Parameters used for activation of video quality improvement for ViNR

Parameter Name	Parameter ID	Option	Setting Notes
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp.*QciAlgoSwitch***	DL_RETRANS_ADJ_OPT_SW	Select this option if PDSCH HARQ retransmission optimization is required according to the network plan.
Retrans CCE Aggregation Level Offset	**gNBDUQciAlgoParamGrp.*RetransCceAggLevelOfs***	None	Set this parameter according to the network plan.
ViNR DL Initial Transmission MCS Decrease	**NRDUCellDlSch**.***VinrDlInitTransMcsDecrease***	None	Set this parameter according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	UL_IBLER_QCI_SW	Select this option if QCI-specific scheduling based on the uplink target IBLER is required according to the network plan.
Schedule Option Switch	**gNBDUSchParamGroup**.***ScheduleOptSwitch***	DL_IBLER_QCI_SW	Select this option if QCI-specific scheduling based on the downlink target IBLER is required according to the network plan.
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	DL_RLC_TIME_DOM_DUPLICATION_SW	Select this option if downlink RLC time-domain duplication is required according to the network plan.
Service Differentiation Algorithm Switch	**NRCellQciBearer**.***ServiceDiffAlgoSwitch***	DL_RESID_ERR_TIME_DOM_DUP_SW	Select this option if time-domain duplication upon downlink residual block errors is required according to the network plan.
UE PDCP Reordering Timer	**gNBPdcpParamGroup**.***UePdcpReorderingTimer***	None	It is recommended that this parameter be set to **MS100** when downlink RLC time-domain duplication is enabled.

Parameter Name	Parameter ID	Option	Setting Notes
Service Differentiation Switch	**NRDUCellAlgoSwitch**.***ServiceDiffSwitch***	CONT_PKT_LOSS_EXIT_TIME_DUP_SW	Select this option if exit from time-domain duplication upon consecutive packet loss is required according to the network plan.
Time-Dom Dup Exit Continuous Pkt Loss Num Thld	**NRDUCellVoiceAlgo**.***TimeDupExtContPktLossThld***	None	Set this parameter according to the network plan.
ViNR DL Jitter Eliminate Duration	**NRCellServExp**.***VinrDUitterEliminateDur***	None	It is recommended that this parameter be set to the same value as the **NRDUCellServExp**.***VonrDUitterEliminateTime*** parameter. You are also advised to use the same value for cells providing contiguous coverage. Otherwise, negative impacts will occur, causing video freezing.

Parameter Name	Parameter ID	Option	Setting Notes
ViNR Downlink Scheduling Switch	**NRDUCellDlSch**.***VinrDlSchSwitch***	VINR_DL_SCH_OPT_DUR_HO_SW	Select this option if downlink scheduling optimization during handovers for ViNR UEs is required according to the network plan. In addition, you are advised to set the **NRDUCellServExp**.***VonrDlJitterEliminateTime*** parameter to a non-zero value to ensure optimal performance.

Table 4-9 Parameters used for activation of mobility management for ViNR

Parameter Name	Parameter ID	Option	Setting Notes
QCI Algorithm Switch	**gNBQciAlgoParamGrp**.***QciAlgoSwitch***	QLTY_BASED_INTER_FREQ_HO_SW	Select this option if video-quality-based inter-frequency handover is required according to the network plan.
Service PLR Evaluation Period	**gNBQciAlgoParamGrp**.***ServPlrEvalPeriod***	None	Retain the default value.
Service Quality Inter-Freq HO PLR Thld	**gNBQciAlgoParamGrp**.***ServQltyInterFreqHoPlrThld***	None	Set this parameter to a value greater than or equal to the **gNBQciAlgoParamGrp**.***ServQltyRecoveryPlrThld*** parameter value according to the network plan.

Parameter Name	Parameter ID	Option	Setting Notes
Service Quality Recovery PLR Threshold	**gNBQciAlgoParamGrp.*ServQltyRecoveryPlrThld***	None	Set this parameter to a value smaller than the **gNBQciAlgoParamGrp.*ServQltyInterFreqHoPlrThld*** parameter value according to the network plan.
Voice and Video Quality Frequency Priority	**NRCellFreqRelation.*VonrQualityFreqPriority***	None	Set this parameter according to the network plan.
Service Quality Inter-Freq HO A5 RSRP Thld 2	**NRCellServExp.*VonrQltyInterFA5RsrpThld2***	None	Retain the default value.
Service Quality Inter-Freq HO A5 RSRQ Thld	**NRCellServExp.*VonrQltyInterFA5RsrqThld***	None	Retain the default value.
Service Quality Inter-Freq HO A5 SINR Thld	**NRCellServExp.*VonrQltyInterFA5SinrThld***	None	Retain the default value.
Inter-Frequency Handover Switch	**NRCellAlgoSwitch.*InterFreqHoSwitch***	INTER_FREQ_HO_COLLABORATION_SW	It is recommended that this option be selected if unnecessary inter-frequency handovers are enabled together with video-quality-based inter-frequency handover.
Inter-Frequency Handover Switch	**NRCellAlgoSwitch.*InterFreqHoSwitch***	VOICE_INTER_FREQ_HO_COLLAB_SW	It is recommended that this option be selected if service-based inter-frequency handover is enabled together with video-quality-based inter-frequency handover.

Parameter Name	Parameter ID	Option	Setting Notes
Voice Strategy Switch	**NRCellAlgoSwitch**.***VoiceStrategySwitch***	QLTY_BASED_REDI_SW	Select this option if video-quality-based redirection is required according to the network plan.

Table 4-10 Parameters used for activation of collaboration with other functions

Parameter Name	Parameter ID	Option	Setting Notes
Service Differentiation Algorithm Switch	**NRDUCellQciBearer**.***ServiceDiffAlgoSwitch***	UL_MU_MIMO_SW	Retain the default value.
DL MU-MIMO Scheduling Supplement Switch	**NRDUCellDlMimo**.***DLMuMimoSchSupplementSw***	VINR_DL_MUMIMO_PROHIBIT_SW	Retain the default value.
QCI Algorithm Switch	**gNBDUQciAlgoParamGrp**.***QciAlgoSwitch***	DRX_WAKEUP_PROTECT_SW	Select this option if DRX wakeup protection is required according to the network plan. The **NRDUCellQciBearer**.***Qci*** parameter can be bound to the **gNBDUQciAlgoParamGrp**.***QciAlgoParamGroupId*** parameter only when the former is set to **1**, **2**, **65**, or **66**.

4.4.1.2 Using MML Commands

Before using MML commands, refer to **4.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- Basic ViNR functions

```
//Enabling basic ViNR functions
MOD NRCELLALGOSWITCH: NrCellId=1, VonrSwitch=VONR_SW-1;
```

- Addition of QCI algorithm parameter groups

```
//Adding a gNodeB DU QCI algorithm parameter group and mapping it to 5QI 2 bearers for bearer-level parameter configuration
ADD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1;
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2, QciAlgoParamGroupId=1;
//Adding a gNodeB QCI algorithm parameter group and mapping it to 5QI 2 bearers for bearer-level parameter configuration
ADD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1;
MOD NRCELLQCIBEARER: NrCellId=1, Qci=2, QciAlgoParamGroupId=1;
```

- Coverage improvement for ViNR

```
//Enabling MAC CE-based rate adjustment
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2, ServiceDiffAlgoSwitch=ANBR_SW-1;
//Setting the ANBR mode (using rate adjustment in the uplink as an example)
MOD GNBQCIBEARER: Qci=2, AnbrMode=UL_ONLY;
//Configuring the ANBR interval
MOD GNBQCIBEARER: Qci=2, AnbrInterval=50;

//Enabling downlink power compensation
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=DL_SERV_EXP_IMP_SW-1;

//Enabling increased downlink retransmissions and downlink fixed rank-1 scheduling
MOD NRDUCELLDLSCH: NrDuCellId=1, VinrDlSchSwitch=VINR_DL_SERV_EXP_IMP_SW-1;

//Enabling enhanced uplink waveform adaptation
MOD NRDUCELLCHNCOVALGO: NrDuCellId=1, UlCoverageAlgoSwitch=RANK1_ADAPT_DFT_SW-1;
```

- Video quality improvement for ViNR

```
//Enabling PDSCH HARQ retransmission optimization
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1,
QciAlgoSwitch=DL_RETRANS_ADJ_OPT_SW-1;

//Configuring the PDCCH CCE aggregation level offset for retransmission
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, RetransCceAggLevelOfs=1;

//Configuring the amount to decrease individual downlink MCS indexes for the initial transmission to ViNR UEs
MOD NRDUCELLDLSCH: NrDuCellId=1, VinrDlInitTransMcsDecrease=1;

//Enabling QCI-specific scheduling based on uplink and downlink target IBLERs
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_IBLER_QCI_SW-1&DL_IBLER_QCI_SW-1;

//Enabling downlink RLC time-domain duplication
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2,
ServiceDiffAlgoSwitch=DL_RLC_TIME_DOM_DUPLICATION_SW-1;
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=1, UePdcpReorderingTimer=MS100;

//Enabling time-domain duplication upon downlink residual block errors
MOD NRCELLQCIBEARER: NrCellId=1, Qci=2,
ServiceDiffAlgoSwitch=DL_RESID_ERR_TIME_DOM_DUP_SW-1;

//Enabling exit from time-domain duplication upon consecutive packet loss and configuring the related threshold
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
ServiceDiffSwitch=CONT_PKT_LOSS_EXIT_TIME_DUP_SW-1;
MOD NRDUCELLVOICEALGO: NrDuCellId=1, TimeDupExtContPktLossThld=2;

//Configuring the downlink jitter elimination duration for ViNR
MOD NRCELLSERVEXP: NrCellId=1, VinrDlJitterEliminateDur=MS100;
```



```
//(Optional) Enabling both downlink scheduling optimization during handovers for ViNR UEs and downlink buffer jitter elimination for VoNR UEs to ensure optimal performance
MOD NRDUCELLDLSCH: NrDuCellId=0, VinrDlSchSwitch=VINR_DL_SCH_OPT_DUR_HO_SW-1;
MOD NRDUCELLSERVEXP: NrDuCellId=0, VonrDlJitterEliminateTime=5;
```

- Mobility management for ViNR

```
//Enabling video-quality-based inter-frequency handover
MOD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=QLTY_BASED_INTER_FREQ_HO_SW-1;
//Configuring the packet loss rate evaluation period, packet loss rate thresholds, and RSRP, RSRQ, and SINR thresholds for event A5 related to service-quality-based inter-frequency handovers
MOD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1, ServPlrEvalPeriod=2, ServQltyInterFreqHoPlrThld=5, ServQltyRecoveryPlrThld=1;
MOD NRCELLSERVEXP: NrCellId=1, VonrQltyInterFA5RsrpThld2=-105, VonrQltyInterFA5RsrqThld=-34, VonrQltyInterFA5SinrThld=44;
//Configuring the frequency priority for video quality
MOD NRCELLFREQRELATION: NrCellId=1, SsbFreqPos=7880, VonrQualityFreqPriority=1;
//Enabling inter-frequency handover collaboration
MOD NRCELLALGOSWITCH: NrCellId=1, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-1;
//Enabling the anti-ping-pong function for service-based and video-quality-based inter-frequency handovers
MOD NRCELLALGOSWITCH: NrCellId=1, InterFreqHoSwitch=VOICE_INTER_FREQ_HO_COLLAB_SW-1;

//Enabling video-quality-based redirection
MOD NRCELLALGOSWITCH: NrCellId=1, VoiceStrategySwitch=QLTY_BASED_REDI_SW-1;
```

- Collaboration with other functions

```
//Configuring collaboration between ViNR and MU-MIMO
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2, ServiceDiffAlgoSwitch=UL_MU_MIMO_SW-1;
MOD NRDUCELLDLMIMO: NrDuCellId=1, DLMuMimoSchSupplementSw=VINR_DL_MUMIMO_PROHIBIT_SW-0;

//Enabling DRX wakeup protection
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=DRX_WAKEUP_PROTECT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- Coverage improvement for ViNR

```
//Disabling MAC CE-based rate adjustment
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2, ServiceDiffAlgoSwitch=ANBR_SW-0;

//Disabling downlink power compensation
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=DL_SERV_EXP_IMP_SW-0;

//Disabling increased downlink retransmissions and downlink fixed rank-1 scheduling
MOD NRDUCELLDLSCH: NrDuCellId=1, VinrDlSchSwitch=VINR_DL_SERV_EXP_IMP_SW-0;

//Disabling enhanced uplink waveform adaptation
MOD NRDUCELLCHNCOVALGO: NrDuCellId=1, UlCoverageAlgoSwitch=RANK1_ADAPT_DFT_SW-0;
```

- Video quality improvement for ViNR

```
//Disabling downlink scheduling optimization during handovers for ViNR UEs
MOD NRDUCELLDLSCH: NrDuCellId=0, VinrDlSchSwitch=VINR_DL_SCH_OPT_DUR_HO_SW-0;

//Disabling PDSCH HARQ retransmission optimization
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, QciAlgoSwitch=DL_RETRANS_ADJ_OPT_SW-0;

//Restoring the PDCCH CCE aggregation level offset for retransmission
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1, RetransCceAggLevelOfs=0;

//Restoring the amount to decrease individual downlink MCS indexes for the initial transmission to ViNR UEs
```

```
MOD NRDUCELLDLSCH: NrDuCellId=1, VinrDlInitTransMcsDecrease=0;

//Disabling QCI-specific scheduling based on uplink and downlink target IBLERs
MOD GNBDUSCHPARAMGROUP: ScheduleParamGroupId=1,
ScheduleOptSwitch=UL_IBLER_QCI_SW-0&DL_IBLER_QCI_SW-0;

//Disabling time-domain duplication upon downlink residual block errors
MOD NRCELLQCIBEARER: NrCellId=1, Qci=2,
ServiceDiffAlgoSwitch=DL_RESID_ERR_TIME_DOM_DUP_SW-0;

//Disabling downlink RLC time-domain duplication
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2,
ServiceDiffAlgoSwitch=DL_RLC_TIME_DOM_DUPLICATION_SW-0;
MOD GNBPDCPPARAMGROUP: PdcpParamGroupId=1, UePdcpReorderingTimer=MS180;

//Disabling exit from time-domain duplication upon consecutive packet loss
MOD NRDUCELLALGOSWITCH: NrDuCellId=1,
ServiceDiffSwitch=CONT_PKT_LOSS_EXIT_TIME_DUP_SW-0;

//Restoring the downlink jitter elimination duration for ViNR
MOD NRCELLSERVEXP: NrCellId=1, VinrDlJitterEliminateDur=MS0;
```

- Mobility management for ViNR

```
//Disabling video-quality-based inter-frequency handover
MOD GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1,
QciAlgoSwitch=QLTY_BASED_INTER_FREQ_HO_SW-0;
//Disabling inter-frequency handover collaboration
MOD NRCELLALGOSWITCH: NrCellId=1, InterFreqHoSwitch=INTER_FREQ_HO_COLLABORATION_SW-0;
//Disabling the anti-ping-pong function for service-based and video-quality-based inter-frequency
handovers
MOD NRCELLALGOSWITCH: NrCellId=1, InterFreqHoSwitch=VOICE_INTER_FREQ_HO_COLLAB_SW-0;

//Disabling video-quality-based redirection
MOD NRCELLALGOSWITCH: NrCellId=1, VoiceStrategySwitch=QLTY_BASED_REDI_SW-0;
```

- Collaboration with other functions

```
//Disabling collaboration between ViNR and MU-MIMO
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2, ServiceDiffAlgoSwitch=UL_MU_MIMO_SW-0;
MOD NRDUCELLDLMIMO: NrDuCellId=1,
DLMuMimoSchSupplementSw=VINR_DL_MUMIMO_PROHIBIT_SW-1;

//Disabling DRX wakeup protection
MOD GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1,
QciAlgoSwitch=DRX_WAKEUP_PROTECT_SW-0;
```

- Removal of QCI algorithm parameter groups

```
//Removing a gNodeB DU QCI algorithm parameter group
MOD NRDUCELLQCIBEARER: NrDuCellId=1, Qci=2, QciAlgoParamGroupId=65535;
RMV GNBDUQCIALGOPARAMGRP: QciAlgoParamGroupId=1;
//Removing a gNodeB QCI algorithm parameter group
MOD NRCELLQCIBEARER: NrCellId=1, Qci=2, QciAlgoParamGroupId=65535;
RMV GNBQCIALGOPARAMGRP: QciAlgoParamGroupId=1;
```

- Basic ViNR functions

```
//Disabling basic ViNR functions
MOD NRCELLALGOSWITCH: NrCellId=1, VonrSwitch=VONR_SW-0;
```

4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.2 Activation Verification

Before verifying whether ViNR has been activated, run the **MOD GNB5QICONFIG** or **ADD GNB5QICONFIG** command with the **gNB5qiConfig**.***Nr5qi*** parameter set to **2** and the **gNB5qiConfig**.***CounterObjSubscriptionInd*** parameter set to **YES** to allow 5QI 2 as the measurement object according to the monitoring plan.

After the preceding configurations are complete, you can observe the counters related to 5QI 2 to check whether ViNR has been activated. When the value of any of the counters listed in **Table 4-11** is not 0, ViNR has been activated.

Table 4-11 Cell-level counters for ViNR activation verification

Counter ID	Counter Name
1911820520	**N.QosFlow.Max.Cell5QI**
1911820521	**N.QosFlow.Avg.Cell5QI**
1911820488	**N.User.RRCConn.Active.DuCell5QI.Max**
1911820489	**N.User.RRCConn.Active.DuCell5QI.Avg**
1911820510	**N.QosFlow.Est.Att.Cell5QI**
1911820511	**N.QosFlow.Est.Succ.Cell5QI**
1911831584	**N.QosFlow.FailEst.AMF.Cell5QI**
1911831585	**N.QosFlow.FailEst.RNL.Cell5QI**
1911831586	**N.QosFlow.FailEst.TNL.Cell5QI**
1911837955	**N.QosFlow.ReEst.Succ.Cell5QI**
1911820513	**N.QosFlow.AbnormRel.Cell5QI**
1911820514	**N.QosFlow.AbnormRel.AMF.Cell5QI**
1911831588	**N.QosFlow.AbnormRel.RNL.Cell5QI**
1911831589	**N.QosFlow.AbnormRel.TNL.Cell5QI**
1911831587	**N.PDCP.DL.TrfSDU.RxPackets.Cell5QI**
1911831590	**N.PDCP.DL.TrfSDU.TxPacket.Discard.Cell5QI**
1911833638	**N.QosFlow.AbnormRel.HOFailure.Cell5QI**
1911837880	**N.QosFlow.AbnormRel.RNL.RlcReset.Cell5QI**
1911837956	**N.QosFlow.FailEst.Conflict.Cell5QI.PLMN**
1911837881	**N.QosFlow.FailEst.RlcReset.Cell5QI.PLMN**
1911837882	**N.QosFlow.FailEst.UeNoReply.Cell5QI.PLMN**
1911837883	**N.QosFlow.AbnormRel.RNL.RlcReset.Cell5QI.PLMN**
1911837884	**N.QosFlow.FailEst.RlcReset.Cell5QI**

Counter ID	Counter Name
1911837885	**N.QosFlow.FailEst.UeNoReply.Cell5QI**
1911837946	**N.PDCP.UL.TrfSDU.RxPacket.Loss.Edge.5QI2**
1911837948	**N.PDCP.UL.TrfPDU.RxPackets.Edge.5QI2**
1911842614	**N.HO.InterRAT.N2E.ReEst.Succ.ViNR.PLMN**
1911842616	**N.HO.IntraFreq.ReEst.Succ.ViNR.PLMN**
1911842618	**N.QosFlow.AbnormRel.HOFailure.ViNR.PLMN**
1911842620	**N.QosFlow.ReEst.Succ.ViNR.PLMN**
1911842622	**N.HO.InterFreq.ReEst.Succ.ViNR.PLMN**
1911844820	**N.UL.SCH.ErrTB.Ibler.DuCell5QI**
1911844821	**N.DL.SCH.ErrTB.Ibler.DuCell5QI**
1911844822	**N.DL.SCH.TB.DuCell5QI**
1911844823	**N.UL.SCH.TB.Retrans.DuCell5QI**
1911844824	**N.DL.SCH.TB.Retrans.DuCell5QI**
1911844825	**N.UL.SCH.TB.DuCell5QI**

Table 4-12 lists the counters used to measure the number of duplicated downlink RLC SDUs for services with a specific 5QI. If the values of the following counters are not 0, downlink RLC time-domain duplication and time-domain duplication upon downlink residual block errors have taken effect.

Table 4-12 Counter related to downlink RLC time-domain duplication

Counter ID	Counter Name	Function
1911840499	**N.RLC.DL.DupTrfSDU.RxPackets.DuCell5QI**	Downlink RLC time-domain duplication
1911844799	**N.RLC.DL.DupTrfSDU.RxPackets.DuCell5QI.Enh**	Time-domain duplication upon downlink residual block errors

The number of inter-frequency outgoing handover attempts, inter-frequency outgoing handover executions, and successful inter-frequency outgoing handover executions after video-quality-based inter-frequency handover is enabled can be measured using the counters listed in **Table 4-13**.

Table 4-13 Counters related to video-quality-based inter-frequency handover

Counter ID	Counter Name
1911840424	**N.HO.InterFreq.ViNRQual.PrepAttOut**
1911840422	**N.HO.InterFreq.ViNRQual.ExecAttOut**
1911840421	**N.HO.InterFreq.ViNRQual.ExecSuccOut**
1911840419	**N.HO.InterFreq.ViNRQual.PrepAttOut.PLMN**
1911840423	**N.HO.InterFreq.ViNRQual.ExecAttOut.PLMN**
1911840420	**N.HO.InterFreq.ViNRQual.ExecSuccOut.PLMN**
1911836056	**N.HO.InterFreq.VoNRQual.PrepAttOut.ViNR**
1911836061	**N.HO.InterFreq.VoNRQual.ExecAttOut.ViNR**
1911836051	**N.HO.InterFreq.VoNRQual.ExecSuccOut.ViNR**
1911836055	**N.HO.InterFreq.VoNRQual.PrepAttOut.ViNR.PLMN**
1911836062	**N.HO.InterFreq.VoNRQual.ExecAttOut.ViNR.PLMN**
1911836057	**N.HO.InterFreq.VoNRQual.ExecSuccOut.ViNR.PLMN**

4.4.3 Network Monitoring

KPIs related to video services:

- Video bearer setup success rate = **N.QosFlow.Est.Succ.Cell5QI**/(**N.QosFlow.Est.Att.Cell5QI – N.QosFlow.Est.Att.Cell5QI.EPSFB**) x 100%
- Video service handover success rate = **N.QosFlow.HoOut.Cell5QI**/(**N.QosFlow.AbnormRel.HoOut.Cell5QI** + **N.QosFlow.HoOut.Cell5QI**) x 100%
- Video service drop rate = **N.QosFlow.AbnormRel.Cell5QI**/(**N.QosFlow.NormRel.Cell5QI** + **N.QosFlow.AbnormRel.Cell5QI**) x 100%
- Video packet loss rate
  - Uplink video packet loss rate = **N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**/(**N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI** + **N.PDCP.UL.TrfPDU.RxPackets.Cell5QI**) x 100%
  - Downlink video packet loss rate = **N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**/**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI** x 100%

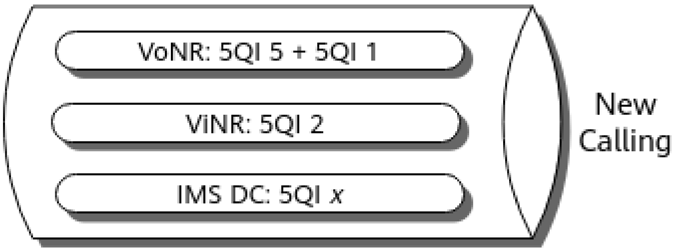
5 New Calling

5.1 Principles

5G New Calling is an advanced communication technology that leverages 5G networks to deliver ultra-high-definition (UHD) voice and video calls, VoNR UHD video calls, 5G video-based customer service, smart translation, content sharing, remote collaboration, augmented reality (AR) calling, and other value-added services. It enables visualized, multimedia, and high-perception UHD communication experiences.

New Calling introduces IMS DCs to the voice and video channels for VoNR and ViNR on IMS networks. IMS DCs allow subscribers to exchange various types of multimedia, such as texts, images, and videos, facilitating visualized, interactive, and immersive communication experiences. **Figure 5-1** shows the relationship among VoNR, ViNR, and New Calling.

Figure 5-1 Relationship among VoNR, ViNR, and New Calling



IMS DC services can be set up on one or more bearers with specific 5QI values, like 9, 71 to 74, and 76, and can be measured using performance counters.

In addition, QoS flow-to-DRB mapping protection for IMS DC services is supported. When an excessive number of QoS flows are set up for IMS DC services of a UE, they can be mapped to at most two guaranteed bit rate (GBR) DRBs and one non-GBR DRB to prevent QoS flow setup failures.

For details about QoS flows, DRBs, and their mapping relationships, see *QoS Management*.

5.1.1 IMS DC Service Measurement Using Performance Counters

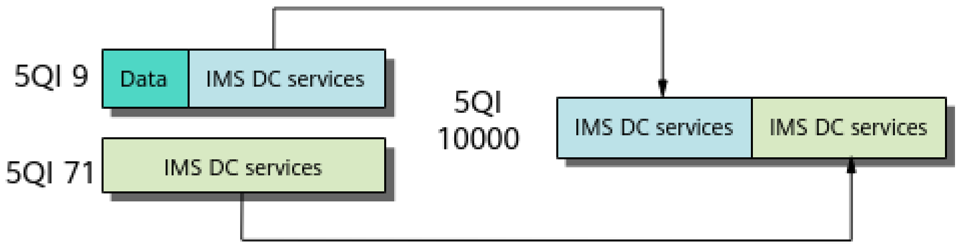
IMS DC services can be set up on one or more bearers with specific 5QI values, like 9, 71 to 74, and 76, and can be measured using performance counters. When the **IMS_DC** option of the **gNB5qiConfig**.***ServiceTypeInd*** parameter is selected, bearers with the corresponding 5QI are dedicated to carrying IMS DC services and will be marked as such in IMS protocol data unit (PDU) sessions.

In RAN sharing with common carrier, bearers carrying IMS DC services can be configured for each operator in a shared cell. When the **IMS_DC** option of the **gNBOp5qiAlgo**.***ServiceTypeInd*** parameter is selected, bearers with the corresponding 5QI are dedicated to carrying IMS DC services and will be marked as such in IMS PDU sessions. For details about multi-operator sharing, see *Multi-Operator Sharing*.

When the **IMS_DC** option of either the **gNB5qiConfig**.***ServiceTypeInd*** or **gNBOp5qiAlgo**.***ServiceTypeInd*** parameter is selected for a specific bearer, the bearer will be identified as carrying IMS DC services, which are measured using performance counters.

In the example shown in **Figure 5-2**, IMS DC services are set up on bearers with 5QIs 9 and 71 for a single operator. When measuring these services using performance counters, the gNodeB associates them with a 5QI 10000 bearer and uses the performance counters related to 5QI 10000. For details about performance counters for IMS DC services, see **5.4.2 Activation Verification**.

Figure 5-2 Example of IMS DC service measurement using performance counters



5.1.2 QoS Flow-to-DRB Mapping Protection for IMS DC Services

When a UE initiates a service, the 5GC maps the service's data flow to one or more QoS flows. Then, the gNodeB maps the QoS flows to DRBs over the air interface. There can be many-to-one or one-to-one mapping relationships between QoS flows and DRBs.

When an excessive number of QoS flows are set up for IMS DC services of a UE, the number of mapped DRBs will exceed the upper limit. This causes failures in QoS flow setup and IMS DC service initiation.

QoS flow-to-DRB mapping protection for IMS DC services can be enabled by selecting the **IMS_DC_QOS_MAP_DRB_PROT_SW** option of the **gNodeBAlgo**.***VoiceSw*** parameter. When an excessive number of QoS flows are set up for IMS DC services of a UE, they can be mapped to at most two GBR DRBs and one non-GBR DRB, thereby increasing the bearer setup success rate of IMS DC services.

5.2 Network Analysis

5.2.1 Benefits

- IMS DC service measurement using performance counters is a basic function that is used to measure the performance of IMS DC services.
- QoS flow-to-DRB mapping protection for IMS DC services increases the QoS flow setup success rate of IMS DC services when an excessive number of QoS flows are set up for IMS DC services of a UE.

QoS flow setup success rate of IMS DC services = **N.QosFlow.Est.Succ.Cell5QI**/(**N.QosFlow.Est.Att.Cell5QI – N.QosFlow.Est.Att.Cell5QI.EPSFB**) x 100%

5.2.2 Impacts

- IMS DC service measurement using performance counters increases the number of subscribed performance counter objects and consumes additional CPU and memory resources for processing performance counters.
- QoS flow-to-DRB mapping protection for IMS DC services may increase the uplink and downlink packet loss rates of IMS DC services and decrease the average uplink and downlink UE throughputs.

In the preceding information:

- Uplink packet loss rate of IMS DC services = **N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**/(**N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI** + **N.PDCP.UL.TrfPDU.RxPackets.Cell5QI**) x 100%
- Downlink packet loss rate of IMS DC services = **N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**/**N.RLC.DL.TrfSDU.RxPackets.DuCell5QI** x 100%
- Average uplink UE throughput = **User Uplink Average Throughput (DU)**
- Average downlink UE throughput = **User Downlink Average Throughput (DU)**

5.3 Requirements

5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-031203	VoNR	NR0S00VONR00	per Cell

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.3.2.1 Prerequisite Functions

None

5.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink QFI compatibility	**DL_QFI_COMPAT_SW** option of the **gNodeBParam**.***CompatibilityAlgoSwitch*** parameter	None	When QoS flow-to-DRB mapping protection for IMS DC services is enabled, the switch controlling downlink QFI compatibility must be turned off.

5.3.2.3 Function Impacts

None

5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.3.4 Cells

- If the **gNB5qiConfig**.***Nr5qi*** parameter is set to **1**, **2**, **5**, **65**, **66**, **67**, **69**, or **70**, the **IMS_DC** option of the **gNB5qiConfig**.***ServiceTypeInd*** parameter must be deselected.
- If the **gNBOp5qiAlgo**.***Nr5qi*** parameter is set to **1**, **2**, **5**, **65**, **66**, **67**, **69**, or **70**, the **IMS_DC** option of the **gNBOp5qiAlgo**.***ServiceTypeInd*** parameter must be deselected.

5.3.5 Networking

SA networking is required.

5.3.6 Others

For details, see **4.3.5 Others**.

5.4 Operation and Maintenance

5.4.1 Data Configuration

5.4.1.1 Data Preparation

Table 5-1 and **Table 5-2** describe the parameters used for function activation.

Table 5-1 Parameters used for activation of IMS DC service measurement using performance counters

Parameter Name	Parameter ID	Option	Setting Notes
Service Type Indicator	**gNB5qiConfig**.***ServiceTypeInd***	IMS_DC	Select this option if bearers with a specified 5QI are required to carry IMS DC services.

Parameter Name	Parameter ID	Option	Setting Notes
Service Type Indicator	**gNBOp5qiAlgo**.***ServiceTypeInd***	IMS_DC	Select this option if bearers with a specified 5QI are required to carry IMS DC services in multi-operator sharing scenarios.
Counter Mapped 5QI	**gNBCounterMap5qiCfg**.***CounterMapped5qi***	None	Set this parameter to **10000** when performance counters related to IMS DC services are subscribed to.
Counter Object Subscription Indicator	**gNBCounterMap5qiCfg**.***CounterObjSubscriptionInd***	NRCELL_5QI	Select this option when subscribing to 5QI-specific counters for IMS DC services in a cell.
Counter Object Subscription Indicator	**gNBCounterMap5qiCfg**.***CounterObjSubscriptionInd***	NRCELL_5QI_OPERATOR	Select this option when subscribing to 5QI-specific counters for IMS DC services of a specific operator in a cell.
Counter Object Subscription Indicator	**gNBCounterMap5qiCfg**.***CounterObjSubscriptionInd***	NRDUCELL_5QI	Select this option when subscribing to 5QI-specific counters for IMS DC services in a DU cell.

Table 5-2 Parameters used for activation of QoS flow-to-DRB mapping protection for IMS DC services

Parameter Name	Parameter ID	Option	Setting Notes
Voice Switch	**gNodeBAlgo**.***VoiceSw***	IMS_DC_QOS_MAP_DRB_PROT_SW	It is recommended that this option be selected if more than two GBR bearers or more than one non-GBR bearer for IMS DC services are planned according to the network plan.

5.4.1.2 Using MML Commands

Before using MML commands, refer to **5.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

- IMS DC service measurement using performance counters
```
//Configuring the service type indicator (using IMS DC services carried on 5QI 9 bearers as an example)
MOD GNB5QICONFIG: Nr5qi=9, ServiceTypeInd=IMS_DC-1;
//Configuring the service type indicator in multi-operator sharing scenarios (using IMS DC services carried on 5QI 9 bearers as an example)
ADD GNBOP5QIALGO: Op5qiConfigIndex=1, OperatorId=0, Nr5qi=9, ServiceTypeInd=IMS_DC-1;
//Configuring the counter object subscription indicator
MOD GNBCOUNTERMAP5QICFG: ServiceType=IMS_DC, CounterMapped5qi=10000,
CounterObjSubscriptionInd=NRCELL_5QI-1&NRCELL_5QI_OPERATOR-1&NRDUCELL_5QI-1;
```
- QoS flow-to-DRB mapping protection for IMS DC services
```
//Enabling QoS flow-to-DRB mapping protection for IMS DC services
MOD GNODEBALGO: VoiceSw=IMS_DC_QOS_MAP_DRB_PROT_SW-1;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

- IMS DC service measurement using performance counters

```
//Restoring the service type indicator (using IMS DC services carried on 5QI 9 bearers as an example)
MOD GNB5QICONFIG: Nr5qi=9, ServiceTypeInd=IMS_DC-0;
//Restoring the service type indicator in multi-operator sharing scenarios (using IMS DC services carried on 5QI 9 bearers as an example)
MOD GNBOP5QIALGO: Op5qiConfigIndex=1, OperatorId=0, Nr5qi=9, ServiceTypeInd=IMS_DC-0;
//Restoring the counter object subscription indicator
MOD GNBCOUNTERMAP5QICFG: ServiceType=IMS_DC, CounterMapped5qi=10000,
CounterObjSubscriptionInd=NRCELL_5QI-0&NRCELL_5QI_OPERATOR-0&NRDUCELL_5QI-0;
```

- QoS flow-to-DRB mapping protection for IMS DC services

```
//Disabling QoS flow-to-DRB mapping protection for IMS DC services
MOD GNODEBALGO: VoiceSw=IMS_DC_QOS_MAP_DRB_PROT_SW-0;
```

5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.4.2 Activation Verification

Observe the counters related to 5QI 10000 to check whether IMS DC services have been activated. If any counter listed in **Table 5-3** has a non-zero value, New Calling has been activated.

Table 5-3 Counters for IMS DC service activation verification

Counter ID	Counter Name
1911820520	**N.QosFlow.Max.Cell5QI**
1911820521	**N.QosFlow.Avg.Cell5QI**
1911820510	**N.QosFlow.Est.Att.Cell5QI**
1911820511	**N.QosFlow.Est.Succ.Cell5QI**
1911820479	**N.PDCP.Vol.UL.TrfPDU.Rx.Cell5QI**
1911820481	**N.PDCP.Vol.DL.TrfPDU.Tx.Cell5QI**
1911820512	**N.QosFlow.NormRel.Cell5QI**
1911820513	**N.QosFlow.AbnormRel.Cell5QI**
1911820515	**N.QosFlow.NormRel.AMF.Cell5QI**
1911820514	**N.QosFlow.AbnormRel.AMF.Cell5QI**
1911820530	**N.QosFlow.HoIn.Cell5QI**
1911820519	**N.QosFlow.Remained.Cell5QI**
1911820534	**N.QosFlow.HoOut.Cell5QI**
1911837996	**N.HO.IntraFreq.IntragNB.PrepAttOut.Cell5QI**

Counter ID	Counter Name
1911837997	**N.HO.InterFreq.IntragNB.PrepAttOut.Cell5QI**
1911837998	**N.HO.IntraFreq.Xn.IntergNB.PrepAttOut.Cell5QI**
1911837999	**N.HO.InterFreq.Xn.IntergNB.PrepAttOut.Cell5QI**
1911838000	**N.HO.IntraFreq.Ng.IntergNB.PrepAttOut.Cell5QI**
1911838001	**N.HO.InterFreq.Ng.IntergNB.PrepAttOut.Cell5QI**
1911838002	**N.HO.IntraFreq.IntragNB.ExecAttOut.Cell5QI**
1911838003	**N.HO.InterFreq.IntragNB.ExecAttOut.Cell5QI**
1911838004	**N.HO.IntraFreq.Xn.IntergNB.ExecAttOut.Cell5QI**
1911838005	**N.HO.InterFreq.Xn.IntergNB.ExecAttOut.Cell5QI**
1911838006	**N.HO.IntraFreq.Ng.IntergNB.ExecAttOut.Cell5QI**
1911838007	**N.HO.InterFreq.Ng.IntergNB.ExecAttOut.Cell5QI**
1911838008	**N.HO.IntraFreq.IntragNB.ExecSuccOut.Cell5QI**
1911838009	**N.HO.InterFreq.IntragNB.ExecSuccOut.Cell5QI**
1911838010	**N.HO.IntraFreq.Xn.IntergNB.ExecSuccOut.Cell5QI**
1911838011	**N.HO.InterFreq.Xn.IntergNB.ExecSuccOut.Cell5QI**
1911838012	**N.HO.IntraFreq.Ng.IntergNB.ExecSuccOut.Cell5QI**
1911838013	**N.HO.InterFreq.Ng.IntergNB.ExecSuccOut.Cell5QI**

5.4.3 Network Monitoring

KPIs related to IMS DC services:

- Bearer setup success rate of IMS DC services = **N.QosFlow.Est.Succ.Cell5QI**/(**N.QosFlow.Est.Att.Cell5QI – N.QosFlow.Est.Att.Cell5QI.EPSFB**) x 100%
- Handover success rate of IMS DC services = **N.QosFlow.HoOut.Cell5QI**/(**N.QosFlow.AbnormRel.HoOut.Cell5QI** + **N.QosFlow.HoOut.Cell5QI**) x 100%
- Service drop rate of IMS DC services = **N.QosFlow.AbnormRel.Cell5QI**/(**N.QosFlow.NormRel.Cell5QI** + **N.QosFlow.AbnormRel.Cell5QI**) x 100%
- Packet loss rate of IMS DC services
  - Uplink packet loss rate of IMS DC services = **N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI**/(**N.PDCP.UL.TrfSDU.RxPacket.Loss.Cell5QI** + **N.PDCP.UL.TrfPDU.RxPackets.Cell5QI**) x 100%

- Downlink packet loss rate of IMS DC services = **N.RLC.DL.TrfSDU.PktUuTrans.Loss.DuCell5QI**/ **N.RLC.DL.TrfSDU.RxPackets.DuCell5QI** x 100%

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

Step 1 Open the EXCEL file of parameter reference.

Step 2 On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

Step 1 Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- *VoNR*
- *SA Mobility Management in Connected Mode*
- *MIMO (TDD)*
- *Turbo Uplink (Low-Frequency TDD)*
- *Massive MIMO Uplink Boosting (Low-Frequency TDD)*
- *URLLC*
- *DRX*
- *High Speed Mobility*
- *License Control Item Lists*
- *Multi-Operator Sharing*
- *QoS Management*
- *Service-differentiated Experience Guarantee*
- *CA Experience Improvement*
- *Uplink Scheduling*

